

Shanmukh Yenikapati

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Senior Software Engineer specializing in agentic AI, production LLM systems, and AI infrastructure. Architected and built an enterprise-scale multi-agent platform from scratch (0-to-1) serving 400K+ weekly visits. Deep experience in RAG, context engineering, and integrating AI workflows into production platforms. Passionate about building foundational AI tools to accelerate software development and operational workflows.

EXPERIENCE

LOWE'S COMPANIES INC

Charlotte, NC

Senior Software Engineer

05/2024 - Present

- One of two core contributors who **architected and built a highly-optimized agentic AI platform from scratch**, orchestrating LLMs and multi-agent workflows for production systems serving **373K+ unique visitors** and **400K+ weekly visits**.
- Designed **multi-agent orchestration architectures** using **Google ADK (production)** and **LangChain/LangGraph** that coordinate specialized agents for automated interactions, tool execution, and complex workflows.
- Engineered **AI infrastructure** and productionized generative AI capabilities, ensuring robust data pipelines, low-latency execution, and seamless integration with existing core APIs.
- Built **RAG systems** and semantic similarity search using **FAISS**, vector indexing, and embedding strategies for enterprise-scale context engineering and fast document retrieval.
- Owned full service lifecycle from design to deployment; platform drove **\$8.1M+ weekly revenue participation** across **20,000+ orders** while maintaining strict system reliability and accountability.

Software Engineer

03/2023 - 05/2024

- Engineered data pipelines with **Apache Airflow, PySpark, and GCP Dataproc** to process and transform daily chatbot data from MongoDB into actionable insights, facilitating advanced data analysis and decision-making.
- Created and maintained comprehensive data analytics dashboards in Apache Superset, producing over **200 visualizations** to monitor user behavior, optimize bot flows, and support strategic adjustments.
- Built an **NLU service** with fine-tuned **BERT** on **FastAPI** to automate operational workflows, improving intent prediction accuracy by **20%**.
- Developed an AI-powered **Transcript Analyzer** using **BERTopic** and OpenAI, extracting actionable insights from large datasets and reducing duplicate topic generation by **40%**.
- Implemented production **ETL pipelines** using **PySpark** and **Apache Airflow** to process large-scale conversation datasets and support AI-driven operational workflows.

WALMART GLOBAL TECH

Bentonville, AR

Software Engineer III Intern

06/2022 - 08/2022

- Detected anomalies in DB insert patterns using SQL-driven statistical models, reducing incident tickets by **25%**.
- Automated validation scripts for misconfigured DB properties, cutting manual resolution efforts by **30%**.
- Optimized ingestion pipelines, achieving **18% latency reduction** in batch jobs.

TATA CONSULTANCY SERVICES

Hyderabad, India

Software Engineer

05/2019 - 07/2021

- Built and scaled backend APIs in **Java Spring Boot** processing millions of **healthcare claims daily** with transaction integrity.
- Developed **Spark pipelines in Databricks** powering real-time dashboards for 300+ agents.
- Migrated PL/SQL → T-SQL on Azure SQL for mission-critical workloads, reducing query latency by **40%**.

PROJECTS

Squadfire | Collaborative Task Management Platform | Python, FastAPI, Postgres, HTMX, Tailwind CSS, Docker | [\[Live\]](#)

- Built end-to-end task management system: async Python backend, realtime UI with HTMX, gamified leaderboards, and automated task scheduling.
- Designed **task scheduling engine** supporting 4 schedule types (daily/interval/weekly/rotating) with auto-generation via cron jobs.
- Implemented **Task Pools** with bulk import and 4 distribution strategies, enabling users to systematically work through 100+ item backlogs.
- Created **analytics middleware** with GeoIP enrichment, device detection, and GDPR-compliant IP anonymization.
- Deployed on **Oracle Cloud VM with Docker, Nginx reverse proxy, SSL**, and CI/CD pipeline via bash scripts.
- Built **email notification service** with async SMTP, **threaded comments** with @mention support, and dynamic leaderboard image generation (Pillow).

Intelligent Chatbot | Python, Tensorflow, NMT, SQLite, Discord | **12/2020**

Developed a Neural Machine Translation-based deep learning model trained on Reddit chat data, integrated a toxicity-filtering score system

IWREN (Improved Wild Relation Network) | Python, Pytorch | **03/2022** | [\[Repo\]](#)

Enhanced an abstract visual reasoning model with architecture updates and dataset augmentation, increasing accuracy by 43%

TECHNICAL SKILLS

- Languages & Frameworks:** Python, Java, Swift, SQL, FastAPI, Spring Boot, React, SwiftUI
- AI & LLMs:** LLMs, Agentic AI, Multi-Agent Systems, RAG, Context Engineering, Google ADK, LangChain, LangGraph, OpenAI, Gemini, FAISS, Semantic Search, BERT
- Infrastructure & Platforms:** Docker, Kubernetes, CI/CD, Google Cloud, Apache Airflow, Dataproc, PySpark, Trino, MongoDB, PostgreSQL, REST APIs

EDUCATION

RUTGERS UNIVERSITY - NEW BRUNSWICK	New Brunswick, NJ
<i>Master of Science in Computer Science; GPA-3.88/4.0</i>	<i>09/2021 - 05/2023</i>
VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY	Hyderabad, India
<i>Bachelor of Technology, Mechanical Engineering; GPA-8.32/10</i>	<i>08/2015 - 05/2019</i>
Relevant Coursework: Data Structures and Algorithms, Operating Systems, Design of Internet Services, Software Engineering Web Applications, Database Management Systems, Artificial Intelligence (AI), Machine Learning (ML), Natural Language Processing (NLP), Machine Vision, Probability and Inference, Brain-Inspired Computing	